Cheat sheet

Agents & MCP

Loops, patterns, tools, memory, protocols and safety

Core

Agent	LLM + tools + memory + control loop + stop condition
ReAct	Thought → Action → Observation loop
Planning	Decompose goal; plan-and-execute; re-plan on failure
Reflection	Critique own output and retry
Memory	Short-term (context), long-term (DB / vectors), episodic (past runs)

Workflow patterns

Prompt chaining	Fixed sequence of steps
Routing	Classify and dispatch to specialist
Parallelisation	Sectioning or voting in parallel
Orchestrator-workers	Lead plans, workers execute subtasks
Evaluator-optimiser	Generate → critique → improve loop

MCP

Host / client / server	App / connector / tool provider
Primitives	Tools (actions), resources (data), prompts (templates)
Transports	stdio (local), streamable HTTP (remote)
Security	Trust only vetted servers; least privilege; review tool permissions

Safety and evaluation

Limits	Max steps, time, tokens, money
Approvals	Human-in-the-loop for irreversible actions
Sandboxing	Run code and browsers in isolated environments
Injection	Treat tool outputs and web content as untrusted
Metrics	Task success, steps, cost, latency, safety incidents, intervention rate